Life table notes-July 2021

health records and information technology (Kenya Medical Training College)

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LIFE TABLE

A life table is an elegant and convenient way of analyzing age specific death rates. Therefore it is a conventional method for summarizing the mortality experiences of any population group, ie it is said to produce or provide concise measures of longevity (long life of period of living) of a particular population. The life table represents a hypothetical cohort of 100,000 persons born at the same instant who experience the rate of mortality represented by qx, the probability that a person age x will die within one year, for each age x throughout their lives. We assume a uniform distribution of deaths for ages greater than 0. For age 0, we use a separation factor, f0, which represents the average number of years not lived by those age 0 who die at age 0. It is one of the effective and compressive method of describing mortality in a population pertaining to age specific death rates, it's considered as a refined and convenient way of analysis. In a simplified and original form it is a representation showing the living and the dying at a particular age thus it is composed of several sets of values showing how a group of infants who were supposed to have been born at the same time and experiencing unchanging mortality conditions would gradually come to an end and die out. William Farr called it a **biometer of population.** Such tables are worked out decennially, after census to show mortality conditions, either during the previous decennium or during a shorter period of covering the date of census. They may be prepared on sex or geographical basis, though the data may not be very common.

USES OF LIFE TABLES

There are as many uses as there are users but to our relevance the following should be noted:-

1. They can be used for studies for population growth.

2. Forecasting the size and distribution of a population at a future date under certain assumptions.

3. Preparation of population congestion by age and sex.

4. Comparison of summarized measures of mortality showing expectation of lives at various ages.

5. Forms the basis of life assurances for the calculation of premia necessary for various amounts of live assurances.

6. Estimation of the size of the future labour force.

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However the accuracy and usefulness of life tables depends mainly upon the accuracy and completeness of registration of births, deaths and the enumeration of the population during census. Shortcoming in death registration are likely to be more than those in the census. Where tables are accurate comparison on mortality can be done on age and international level

CONSRUCTION OF A LIFE TABLE

Many methods and approaches exists. However our main concern will concentrate in a method where satisfaction and required data on births and deaths in a population is available. In constructing life tables one requires to have a visually active mind showing survival and deaths occurring in specified generation of babies or births. One can also ask that what would happen if a specified number of babies e.g 100,000 born at the same time were subjected to mortality influence at various ages that was found to be influencing the population at a certain period of time. To follow this, a specified number of babies in a live table will show how many of these would live to see their first, second, third and …… birth day until the last one. When deaths are known we can be able to obtain mortality which when applied to the specified generation one can be able to estimate the number of those surviving. For example if the mortality rate of the second year survival is applied to the number of surviving at the end of the 1st year, we would be able to estimate the number who survived at second year of life. The knowledge for computation of **age specific mortality rate** is necessary in these calculations.

There are several life table abbreviations or functions which are commonly accepted to enhance the understanding of the construction of a life table. It is necessary to acknowledge these abbreviations which include:-

X – age

lx – living at the age X,

dx – dying between age X and X+1

qx- mortality rate or probability of death between age X and X+1

Px -the survival rate

Lx – living between age X and X+1

Tx – Living above age X

e’x – Complete expectation of life at age X.

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Common life tables consists of 8 columns represented by the above abbreviations. However some may carry more or less depending on the need, type of data and method of computation. It is worth noting that in a given set of columns one can obtain any other column. Given some of the columns. The only raw data involved is the age specific mortality rate. Most tables may be constructed on a one year interval but any other interval (s) exists. The common life tables start at 0-1 years,1-5 years and at the interval of 5 years until the last open ended interval (65+,75+ etc)This is commonly known as ABRIDGED LIFE TABLE. The abbreviations are normally added with an **n** to denote the length or size of the interval.

Life table always start with a specified number of births or population which is referred to as a X calculations. The following descriptions apply for each of these functions:

i) X = This is normally given and it indicates the exact year (s) of age
ii) lX =This is the number of persons who are expected to attain the exact age (X) and of the radix (assumed births)
iii) dX =This is the number of of lX persons who died before reaching age X+1.It is written as dX =LX –LX +1
iv) qX =This is the mortality rate to which the population age group would be exposed throughout their life time. The formular is as follows
dX/LX
v) PX = This gives the probability that a group of the precise age X will survive until his next birthday. Thus a person must either live or die at a particular year.The formular is as follows PX = 1-qX
vi) LX = This gives the number of years lived in aggregated by lX persons between age X and X+1.It gives the distribution of the life table stationery population. This is an assumption that deaths in each year of age are evenly spread throughout).The formula is as follows LX = lX -1/2dX.
vii) TX = this gives the number of years lived by a group from age X until all of them die. The formula is as follows
TX=lx+LX+1+Lx +2----------Ln or Lo +L1+L2----------Ln.

viii) e'x = This is the average number of years a person of age x can be expected to live under the prevailing mortality conditions. It is referred to as LIFE EXPECTANCY or expectation (longevity).The formula is e'x = TX/LX.

A group of 142759 children were born at one time in a given population and the experience of death was recorded as follows:-

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Table 1

X	lx	dX	qX	PX	LX	TX	e'x
0	142759	27124	0.19000	0.81000	129197	4632557	32.45
1	115635	7059	0.06000	0.94000	112105.5		
2	108576	3960	0.04000	0.96000	106596		
3	104616	2610	0.02000	0.98000	103311		
4	102006	2006	0.02000	0.98000	101003		
5	100,000	1702	0.02000	0.98000	99149		
6	98298	1600	0.02000	0.98000	97498		
7	96698	1483	0.02000	0.98000	95956.5		
8	95215	1383	0.01000	0.99000	94523.5		
9	93832	1293	0.01000	0.90000	93185.5		
10	92539	1210	0.01000	0.90000	91934		

The following table shows the top and bottom parts of a life table for California 1970.

Table2

AGE INTERVAL	MID YEAR POP. Px	DEATHS IN THE YEAR Dx	ASDR Mx	FRACTION OF LAST YEAR LIVED ax	PROBABILITY OF DYING qx
0-1	340,483	6234	0.018309	0.09	0.018009
1-2	326.,154	369	0.001128	0.43	0.001127
2-3	313,699	269	0.000858	0.43	0.000858
3-4	323,441	237	0.000733	0.47	0.000733
4-5	338,904	175	0.000516	0.49	0.00516
5-6	382,151	179	0.00494	0.50	0.000494
6-7	379,642	171	0.000450	0.50	0.000450
83-84	34,439	3753	0.108975	0.50	0.103344
84-85	31,009	3669	0.118320	0.50	0.111711

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85+	142,691	22483	0.047564	-	1.000000

Points to note from this example are:

1. Mx and qx are generally similar to each other, with Mx slightly larger in growing populations.
2. The ax used here were calculated from the California registration data and a specific to California 1970.
3. With the last open ended age group, because all those alive at the start of the interval will die before its end, simply because it has no end qx is always 1.000.
4. For qx the infant mortality rate could have been used.
5. This example is always very unusual in that the life table is for both sexes combined. Almost always the sexes are treated separately.

The California life table is an example of a complete life table because single year age groups are used throughout. In demography generally abridged life tables are more common than complete ones.

INTERPRETATION OF LIFE TABLE.

Life table functions has subject to two different applications depending on the interpretations given to a life table as a whole. In the more common interpretation a life table is viewed as showing life time mortality experience of a single cohort of newborn babies who are subject to age specific mortality rate on which the table is based. In the second interpretation of the life table it is viewed as a stationary population resulting from the unchanging schedule of age specific mortality rate shown under constant annual number of births.

The interpretation of the life table as the history of a cohort is applied in public health studies and mortality analysis and calculating survival rates for estimating population, net migration, fertility and reproductivity.The interpretation of a life table as a stationary population is used in the comparative measurements of mortality and in the study of population structure.

Table 3.4 depicts a hypothetical complete life table for India during 1971-80.

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TABLE 3.4
Hypothetical Complete Life Table for India (1971–80)

Age x to $x+n$	dx	fx $fx-dx$ (3–2)	qx dx/fx (2/3)	px $1-qx$ (1–4)	Lx $(fx0+fx1)/2$	Tx $Tx1 = T\times x0 - L\times x0$	ex Tx/Fx
1	2	3	4	5	6	7	8
0.	13000	100000	0.13000	0.87000	93500	4930250	49.3
1.	1300	87000	0.01494	0.98506	86350	4836750	55.6
2.	1200	85700	0.01400	0.98600	85100	4750400	55.4
3.	100	84500	0.00118	0.99882	84450	4665300	55.2
4.	800	84400	0.00948	0.99052	84000	4580850	54.3
5.	500	83600	0.00598	0.99402	83350	4496850	53.8
6.	300	83100	0.00361	0.99639	82950	4413500	53.1
7.	200	82800	0.00242	0.99758	82700	4330550	52.3
8.	100	82600	0.00121	0.99879	82550	4247850	51.4
9.	70	82500	0.00085	0.99915	82465	4165300	50.5
10.	65	82430	0.00079	0.99921	82398	4082835	49.5
11.	60	82365	0.00073	0.99927	82335	4000438	48.6
12..	80	82305	0.00097	0.99903	82265	3918103	47.6
13.	100	82225	0.00122	0.99878	82175	3835838	46.7
14.	200	82125	0.00244	0.99756	82025	3753663	45.7
15.	210	81925	0.00256	0.99744	81820	3671638	44.8
16.	215	81715	0.00263	0.99737	81608	3589818	43.9
17.	220	81500	0.00270	0.99730	81390	3508210	43.0
18.	225	81280	0.00277	0.99723	81168	3426820	42.2
19.	230	81055	0.00284	0.99716	80940	3345653	41.3
20.	225	80825	0.00278	0.99722	80713	3264713	40.4
21.	230	80600	0.00285	0.99715	80485	3184000	39.5
22.	235	80370	0.00292	0.99708	80253	3103515	38.6
23.	240	80135	0.00299	0.99701	80015	3023263	37.7
24.	245	79895	0.00307	0.99693	79773	2943248	36.8
25.	260	79650	0.00326	0.99674	79520	2863475	36.0
26.	275	79390	0.00346	0.99654	79253	2783955	35.1
27.	285	79115	0.00360	0.99640	78973	2704703	34.2
28.	295	78830	0.00374	0.99626	78683	2625730	33.3
29.	320	78535	0.00407	0.99593	78375	2547048	32.4
30.	350	78215	0.00447	0.99553	78040	2468673	31.6
31.	400	77865	0.00514	0.99486	77665	2390633	30.7

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32.	450	77465	0.00581	0.99419	77240	2312968	29.9
33.	550	77015	0.00714	0.99286	76740	2235728	29.0
34.	600	76465	0.00785	0.99215	76165	2158988	28.2
35.	700	75865	0.00923	0.99077	75515	2082823	27.5
36.	800	75165	0.01064	0.98936	74765	2007308	26.7
37.	900	74365	0.01210	0.98790	73915	1932543	26.0
38.	1000	73465	0.01361	0.98639	72965	1858628	25.3
39.	1050	72465	0.01449	0.98551	71940	1785663	24.6
40.	1150	71415	0.01610	0.98390	70840	1713723	24.0
41.	1200	70265	0.01708	0.98292	69665	1642883	23.4
42.	1300	69065	0.01882	0.98118	68415	1573218	22.8
43.	1325	67765	0.01955	0.98045	67103	1504803	22.2
44.	1375	66440	0.02070	0.97930	65753	1437700	21.6
45.	1400	65065	0.02152	0.97848	64365	1371948	21.1
46.	1450	63665	0.02278	0.97722	62940	1307583	20.5
47.	1475	62215	0.02371	0.97629	61478	1244643	20.0
48.	1500	60740	0.02470	0.97530	59990	1183165	19.5
49.	1525	59240	0.02574	0.97426	58478	1123175	19.0
50.	1550	57715	0.02686	0.97314	56940	1064698	18.4
51.	1600	56165	0.02849	0.97151	55365	1007758	17.9
52.	1625	54565	0.02978	0.97022	53753	952393	17.5
53.	1650	52940	0.03117	0.96883	52115	898640	17.0
54.	1675	51290	0.03266	0.96734	50453	846525	16.5
55.	1700	49615	0.03426	0.96574	48765	796073	16.0
56.	1725	47915	0.03600	0.96400	47053	747308	15.6
57.	1750	46190	0.03789	0.96211	45315	700255	15.2
58.	1755	44440	0.03949	0.96051	43563	654940	14.7
59.	1760	42685	0.04123	0.95877	41805	611378	14.3
60.	1765	40925	0.04313	0.95687	40043	569573	13.9
61.	1760	39160	0.04494	0.95506	38280	529530	13.5
62.	1750	37400	0.04679	0.95321	36525	491250	13.1
63.	1725	35650	0.04839	0.95161	34788	454725	12.8
64.	1700	33925	0.05011	0.94989	33075	419938	12.4
65.	1675	32225	0.05198	0.94802	31388	386863	12.0
66.	1650	30550	0.05401	0.94599	29725	355475	11.6
67.	1600	28900	0.05536	0.94464	28100	325750	11.3
68.	1575	27300	0.05769	0.94231	26513	297650	10.9
69.	1550	25725	0.06025	0.93975	24950	271138	10.5
70.	1500	24175	0.06205	0.93795	23425	246188	10.2

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The above **life table** provides the column wise information which is generally provided and followed by all life tables.

Col. 1.x = Specific Age

If the age at birth is x then the age at one year is x + 1. Similarly the age at 15 years is x + 15.

Col. 2. dx = Number of deaths, at any particular age. i.e., at the age x, 13000 deaths occur out of 100,000 births, then at age x + 1: 87,000 persons will be alive. In this age, if 1300 deaths occur then at age x + 2: 85700 persons will be alive.

Col. 3. fx = The number of persons surviving at age x to x + n i.e., at the age x + 1 = 1,00,000- 13,000 = 87,000

Col. 4. qx = Probability of death per person in the specific age i.e., total deaths occurred. (Out of 100,000 = 13,000)

Probability = 13,000÷ 100,000 = 0.13

Similarly, at the age x + 1, 1300 persons died out of 87,000 live population then

Probability = 1300 ÷ 87,000 = 0.01494.

Col. 5. Px = 1 – Probability of surviving per individual person or 1 – qx, i.e.;

At age x, 1 – .13000 = .87000 and at age x + 1, 1- .01494 = .98506.

Col. 6. Lx = Number of years lived by the cohort in the age x to x + n or

fx of any two age groups ÷ 2

Suppose,

fx- 15 = 81925 [col.: 3, row 15]

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fx- 16 =81715 [col.: 3, row 16]

Lx = 163640 ÷ 2 = 81820

Col. 7. Tx – Total number of years lived by the cohort after exact age x.

This can be found out from the reverse side of life table, i.e.,

At the age of 94 Lx = 525 and

at age 93 Lx = 925

then at age x + 93, total number of years lived by

Cohort = 525 + 925 = 1450 and at age x + 92,

it will be 525 + 925 + 1400 = 2850.

Col. 8. Ex = T.v ÷Fx. This gives average life expectancy.

In short, the life table is based on the **age of death period** of a particular population. By studying this table, we can show the **probability of death** of any person of any particular age group. It is to be noted that every person of a particular age group does not die according to the estimate of the life table. The life table only shows a trend. After studying the table above, you can see how various formula can be arrived at for future reference in case you come across a life table question requiring you to fill in blank columns eg **ex(Life expectancy at age x)=TX/FX (**Total number of years lived by the cohort after exact age x/ The number of persons surviving at age x to x + n). **Regular practice is key to familiarize yourselves with the formula.**

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Life-tables and their demographic applications

One important method of assessing the health of a population is to ask how long people can expect to live. Life expectancy, usually reported at birth although it can be applied to other ages as well, is a commonly used summary measure which can also be used to compare against countries. Life expectancy is calculated using life tables.

Life tables

A life table is a table which shows, for a person at each age, what the probability is that they die before their next birthday. From this starting point, a number of statistics can be derived and thus also included in the table is:

- the probability of surviving any particular year of age
- the remaining life expectancy for people at different ages
- the proportion of the original birth cohort still alive.

Life tables are usually constructed separately for men and for women because of their substantially different mortality rates.

Life tables are also used in biology.

Construction of life tables

Age specific mortality rates are applied to a notional population, typically of 100,000. Starting at birth, the probability of dying in each period is applied to the number of people surviving to the beginning of the period, so that the initial figure slowly reduces to zero. The different elements required for a life table include (using standard notations):

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Ix	Number of survivors at age x
nqx	Probability of dying between age x and x+n
nDx	Number of deaths between age x and x+n
nLx	Number of person years lived between age x and x+n
Tx	Total number of person years lived after age x
ex	Life expectancy at age x

This sort of life table is based on current age-specific death rates for each age or age band used and are called period life tables and are the most frequently used type. In contrast, actual life expectancy of a particular birth cohort can only be calculated when everyone in this cohort is dead. This approach uses a cohort life table and requires data over many years to prepare just a single complete cohort life table.

Example

An example of how a life table can be constructed and the mathematics involved is provided above.

Strength

Summary measure of mortality providing an overall picture of mortality, allowing countries and regions to be compared.

Weaknesses

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Hypothetical measure that has the potential to be misunderstood by general public/media.

It does not say much about who is still alive, and their quality of life; for example, how many years are lived with disability before dying. This has led to attempts to bring together morbidity and mortality, with measures such as Health Adjusted Life Expectancy and Disability Adjusted Life Years.

Health Adjusted Life Expectancy (HALE)

This is calculated by subtracting from the life expectancy a figure which is the number of years lived with disability multiplied by a weighting to represent the effect of the disability.

If A = years lived healthily

B = years lived with disability

A+B = life expectancy

A+*f*B = healthy life expectancy, where *f* is a weighting to reflect disability level.

N.B. This raises all sorts of moral questions on who defines and measures disability level and how they do it.

Disability Adjusted Life Years (DALY)

Conversely, DALYs combine death and years lived with disability to estimate the burden of disease on populations, and DALYs were used in the Global Burden of Disease study to enable mortality and morbidity comparisons across countries. Weightings were applied to conditions by using the time trade off approach, in which people were asked to consider living more years in imperfect health compared with fewer years in perfect health. The study also placed more weight on the life of a young adult compared with a new born.

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Potential Years of Life Lost (PYLL)

Method for calculating YPLL from a line listing

Step 1. Decide on end point (65 years, average life expectancy, or other).

Step 2. Exclude records of all persons who died at or after the end point.

Step 3. For each person who died before the end point, calculate that person's YPLL by subtracting the age at death from the end point.

$YPLL_{individual}$ = end point – age at death

Step 4. Sum the individual YPLLs.

$YPLL = \sum YPLL_{individual}$

Method for calculating YPLL from a frequency

Step 1. Ensure that age groups break at the identified end point (e.g., 65 years). Eliminate all age groups older than the endpoint.

Step 2. For each age group younger than the end point, identify the midpoint of the age group, where midpoint = age group's youngest age in years + oldest age + 1

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Step 3. For each age group younger than the end point, identify that age group's YPLL by subtracting the midpoint from the end point.

Step 4. Calculate age-specific YPLL by multiplying the age group's YPLL times the number of persons in that age group.

Step 5. Sum the age-specific YPLLs.

The **YPLL rate** represents years of potential life lost per 1,000 population below the end-point age, such as 65 years. YPLL rates should be used to compare premature mortality in different populations, because YPLL does not take into account differences in population sizes.

The formula for a YPLL rate is as follows:

Years of potential life lost

x 10n

Population under age 65 y

A measure related to HALE and DALY, this measure attempts to quantify the potential years of life lost by looking at average age of death from conditions compared to average life expectancy. PYLL can be expressed absolutely or as a rate relative to the population at

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risk.

Other applications

Other characteristics can also be used to distinguish different risk factors for life expectancy, such as smoking-status, occupation, socio-economic class, and others. More complex analyses for assessing cancer survival, that involves comparisons between two populations or a population in two points in time can also be undertaken.

In addition to public health domains, life tables are also used by insurance companies and actuary departments.

When used in biology, age specific fertility rates are also included in the calculations.

When data have not been available, such as in low income countries, life tables have been modelled using what data are available, usually childhood mortality data.

Assignment

In this exercise you learn to:

1. Build a hypothetical life table and

2. Understand how modifications to the life table can generate different fitness outcomes.

You see an **Abridged Life Table** with six variables: Age Category, Death Rate, Survival Rate, Survivorship, Fertility, and Reproductive Value. The only variables you will modify in this exercise are Death Rate and Fertility. In terms of fitness, only Mortality and Fertility really matter.

Enter the Mortality Rates and Fertility Rates according to the outline below.

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Age	Mortality Rate	Survival Rate	Survivorship	Fertility
0 - 9	0.15			0
10 - 19	0.08			1
20 - 29	0.03			3
30 - 39	0.03			3
40 - 49	0.05			1
50 - 59	0.1			0
60 - 69	0.3			0
70 - 79	0.45			0
80 - 89	0.55			0
90 - 99	0.75			0
100+	1			0
		Reproductive Value:		

Survival rate is the chance of dying during an age interval. Calculate survival rate as 1 minus the Mortality Rate. Enter this calculation into the Survival Rate column.

Survivorship is the chance of surviving from birth until the BEGINNING of the age category in question. Set Survivorship to the first age class equal to 1. This means everyone survives to birth.

The Survivorship for all subsequent age classes equals the chance of surviving during the previous age class multiplied by the survivorship to the previous age class. This may seem strange. For example, the chance of surviving to your 30th birthday equals the chance of surviving to your 20th birthday multiplied by the chance of your surviving the age interval of 20 to 29 years.

As a check, make sure Survivorship gets smaller as age class increases. Logically, you can't have a greater chance of surviving from birth to 30 than of surviving from birth to 20.

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Now calculate Reproductive Value. Recall that Reproductive Value represents the number of babies that an average person will have in a lifetime. This equals the product of survivorship and fertility summed over all ages.

Life table practice

Below are five **self-assessment quizzes** to test your understanding of life tables. All the calculations are based on the figures shown in the table below (Table 3.4 as shown previously).

age x	l_x	age x	l_x	age x	l_x	age x	l_x
0	100 000	20	99 277	45	97 817	70	85 201
1	99 562	25	99 148	50	96 923	75	77 504
5	99 484	30	98 981	55	95 510	80	65 457
10	99 440	35	98 743	60	93 393	85	48 163
15	99 387	40	98 386	65	90 159		

Q1. What is infant mortality per thousand in the table above? Express your answer to 2 decimal places, followed by the phrase per thousand, e.g. 1.23 per thousand.

Q2. What is Under 5 mortality per thousand in the table? Express your answer to 2 decimal places, followed by the phrase per thousand, e.g. 1.23 per thousand.

Q3. What is the probability of surviving between exact ages 30 and 40 in the table? Express your answer to 3 decimal places or one decimal place of a percentage (%) e.g. 0.999 or 99.9%

Q4. What is the probability of dying between 30 and 40 as in the last question? Express your answer in 3 decimal places or a decimal place of a percentages (%) e.g. 0.002 or 0.2%

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